

Eric Guo

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EDUCATION

University of Toronto

Honours Bachelor of Science in Computer Science and Statistics

Toronto, ON

Sept. 2021 – May 2026

TECHNICAL SKILLS

Languages: Java, Python, C, C++, SQL, JavaScript, TypeScript, Bash

Frameworks: FastAPI, Starlette, Node.js, Next.js, Dagster, PyTest, JUnit

Cloud/Infra: Kubernetes, Docker, AWS S3, GCP, OpenShift, Helm, GitHub Actions, GitLab CI

Data: Apache Arrow, Parquet, Pandas, ETL, Data Modeling, MapReduce, SLURM, PostgreSQL, MongoDB, DuckDB

Skills: Backend Systems, Data Structures & Algorithms, Microservices, Distributed Data Processing, Object-Oriented Programming, Cloud Infrastructure, AI Tools (Codex, Claude, Copilot)

EXPERIENCE

Machine Learning Software Engineer

May 2025 – Dec 2025

RBC Borealis

Toronto, Ontario

- Designed and tested end-to-end **Dagster ETL pipelines** using a **map-reduce** pattern to process **terabytes** of client embedding data, storing outputs in **AWS S3** as **Apache Arrow** and **Parquet** formats.
- Built production-ready **REST APIs** with **FastAPI**, integrating OAuth 2.0 to serve embeddings in under 100ms on **OpenShift** and **GCP**; improved 99th percentile performance by **400%** through **horizontal resource scaling**.
- Developed **CI/CD pipelines** in GitHub Actions, deploying microservices to **OpenShift** clusters, leveraging **layering** and **caching** to reduce Docker image size by up to 20GB and build times by 15 minutes.
- Conducted **load testing** and **unit testing** with **Postman** and **PyTest**, writing **scalable, async** software.
- Leveraged **Copilot** and **Claude** to build **well-documented** software components quickly, ensuring production readiness through peer and mentor code reviews.

Network Automation

Apr 2024 – Aug 2024

NOKIA Corp.

Ottawa, Ontario

- Developed **CI/CD pipelines** in GitLab CI to run unit tests and deploy product to **Kubernetes**.
- Achieved 97% test coverage by using a **test-driven development** approach with **PyTest**, including unit tests for **regex**-based validation logic.
- Migrated** a company product from a **Docker Compose** deployment to **Kubernetes** on Rancher using **Helm charts**, improving deployment automation and reducing manual operational effort.

Network Automation

Apr 2023 – Aug 2023

NOKIA Corp.

Ottawa, ON

- Tested** and **debugged** Network Automation Platform (NAP) by simulating deployments in **Linux VMs**.
- Programmed custom scripts using **Python** w/ **Pandas** and **NumPy** to handle client-specific configuration uploads, resulting in more streamlined and error-free automation.
- Utilized **Postman** to test **RESTful API** functionality for uploading and retrieving configurations, ensuring seamless communication with **MongoDB**.

Software Engineer Intern

May 2022 – Aug 2022

LUCIAI Inc.

Remote

- Migrated IoT data from NoSQL database, **MongoDB**, to **relational database PostgreSQL** using **Node.js**, **Mongoose**, and **Prisma** for scalability, improving database efficiency by 30%.
- Developed and tested **RESTful APIs** using **JavaScript** and **Postman**, validating API workflows and ensuring system reliability.

PROJECTS

Scriptorium | Node.js, Next.js, TypeScript, React

Sept. 2024 – Jan. 2025

- Developed a code execution platform using **Next.js**, supporting multi-language execution, syntax highlighting, and secure **Docker**-based code execution with timeout controls and isolated runtime environments.
- Built and tested backend **REST APIs** with **JWT** authentication, validating workflows across multiple endpoints and documenting them with **Swagger UI**.
- Designed a relational **SQL** schema for users, blogs, and templates to support fast, scalable data retrieval.
- Implemented responsive frontend features for mobile and cross-browser users using **React** and **Tailwind CSS**.